

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Detailed Marking Report — Chemistry Exam-19

Seemab · Class 9 (Science Group) · Revision Test, Chapters 1 to 9 · Attempted 21 August 2026

TOTAL SCORE 35 / 43	PERCENTAGE 81.4%	GRADE A	SECTION A 17 / 24	SECTION B 9 / 9	SECTION C 9 / 10
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How this paper was marked. Seemab wrote her MCQ answers as a numbered list on the first page rather than filling the bubble grid, which is her usual habit. Her Section A score of 17/24 sits well below her established baseline, so under the standing rule every one of the 24 answers was re-read from the scan at 400 DPI before any mark was recorded, and the answer key was independently re-extracted from the published PDF rather than taken from memory. The printed option order for all seven of her wrong answers was then checked against the paper itself. **Every read was confirmed, the key matched, and all seven questions sit on stated Student Learning Outcomes.** Nothing on this paper needed voiding, unlike Physics exam-13 and Chemistry exam-18.

SECTION A — Multiple Choice (17 / 24)

#	Topic	Hers	Key	Mark	Comment and source in her textbook
1	Definition of chemistry	B	B	1	Correct. Ch1 SLO 1.
2	Most compressible state	C	C	1	Correct, gas. Ch2 §2.1 macroscopic properties.
3	Which is a compound	C	C	1	Correct, carbon dioxide. Ch2 SLO.
4	Saturated solution	A	A	1	Correct. Ch2 §2.5.
5	Strong nuclear force	D	D	1	Correct, and this sub-topic had never been examined before. Ch3 SLO.
6	Shell number vs energy and distance	A	B	0	She chose "both decrease". A larger shell means higher energy and a greater average distance from the nucleus. Ch3 SLO (shells as energy levels). Valid question.
7	Nucleon number	A	A	1	Correct, protons + neutrons. Ch3 SLO.
8	Isotopes	C	C	1	Correct. Ch3 SLO.
9	Order of the periodic table	B	B	1	Correct, proton number. Ch4 SLO 1.
10	Group 2 ion charge	D	D	1	Correct, +2. Ch4 SLO.
11	Atomic radius down a group	A	A	1	Correct, increases. Ch4 SLO (trends).
12	Group 17 name	C	C	1	Correct, halogens. Ch4 SLO.
13	Metallic bond	A	B	0	She chose "two oppositely charged ions", which is the ionic bond. A metallic bond is between a sea of delocalized electrons and metal cations. Ch5 SLO. She defined the ionic bond correctly in Section B , so this is a mix-up of which bond was asked for, not ignorance of the definition.
14	Metals electropositive	A	D	0	She chose "gain electrons". Electropositive means metals lose electrons to form positive ions. Ch5 SLO.
15	Ionic vs covalent melting points	C	A	0	She chose "no melting point at all". Ionic compounds have higher melting points because of the strong lattice forces. Ch5 SLO.
16	Configuration atoms aim for	A	C	0	She chose "alkali metal". Atoms bond to reach the nearest noble gas configuration, which is the basis of the octet rule. Ch5 SLO.
17	Empirical formula meaning	D	D	1	Correct. Ch6 SLO.
18	Moles in 80 g NaOH	B	B	1	Correct, 2 mol. Ch6 SLO.
19	State symbol (aq)	A	A	1	Correct, dissolved in water. Ch6 SLO. <i>Worth noting against Q.3(i)(c) below.</i>
20	Formula from Ca^{2+} and NO_3^-	B	D	0	She chose Ca_2NO_3 , which puts the subscript on the metal. One Ca^{2+} needs two nitrates: $\text{Ca}(\text{NO}_3)_2$, brackets required. Ch6 SLO.
21	Balancing $\text{Zn} + \text{O}_2$	B	B	1	Correct, 2, 1, 2. Ch6 SLO.

22	Free-state oxidation number	C	C	1	Correct, zero. Ch7 SLO. First time this was examined.
23	Monoatomic ion oxidation number	A	A	1	Correct, and she had crossed out a first answer to get here. Ch7 SLO.
24	Activation energy	B	D	0	She chose "change of physical state". It is the minimum energy for a successful collision . Ch8 SLO.

The pattern in Section A. Four of her seven losses (Q13, Q14, Q15, Q16) are the **entire Chapter 5 block on chemical bonding**. She answered every other chapter well: Ch1 1/1, Ch2 3/3, Ch3 3/4, Ch4 4/4, Ch6 4/5, Ch7 2/2. This is not a general dip, it is one chapter. **Chapter 5 is the single most valuable thing for her to revise.**

SECTION B — Short Questions (9 / 9)

Q.2(i) Formation of calcium fluoride, CaF_2 3 / 3

Her dot-and-cross diagram is correct. She drew calcium with its shells, marked it **+2**, and drew curved arrows carrying one electron to each of **two** separate fluorine atoms, each bracketed and marked **-1**. The one-to-two ratio was the discriminator in this question and she got it right. Full marks for (a).

For (b) she wrote:

"Ionic bond : a bond formed by the electrostatic attraction attraction b/w oppositely charged ions."

Correct definition, so the mark stands. She wrote "attraction" twice and spelled it "attraction"; no mark is affected. **Note the contrast with MCQ 13**, where she gave this same definition as the answer for a metallic bond.

Q.2(ii) Relative atomic mass of lithium 3 / 3

(a) She set out $(6 \times 7.5 + 7 \times 92.5) \div 100 = 692.5 \div 100 = \mathbf{6.925}$. Correct, with the formula shown. **2**

(b) Protons 3, electrons 3, neutrons 4 for ${}^7\text{Li}$. Correct. **1**

Q.2(iii) Beryllium, $Z = 4$ 3 / 3

(a) $1s^2 2s^2$, correct, and she added block = s. **1**

(b) "Group (2) because beryllium has 2 electron it in it's shell" and "Period (2) because it has 2 occupied electron shells". Both reasons correct. **1**

(c) She wrote $\text{Be} \rightarrow \text{Be}^{2+} + 2e^-$ and explained:

"Since Be in group 2, it has 2 electron in it's valence shell so it tends to lose both 2 electron and forms 2+ ions."

Exactly the reasoning the question wanted, linking group number to ion charge. **1**

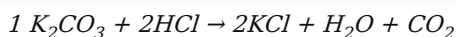
SECTION C — Long Questions (9 / 10)

Q.3(i) Empirical and molecular formula 4 / 5

(a) Definitions 1 / 1 — both correct, empirical as "the simplest whole number ratio of atoms of each element in a compound" and molecular as "the actual number of atom of each element in one molecule of a compound".

(b) The calculation 3 / 3 — fully correct and cleanly laid out. She divided by the relative atomic masses (1.449, 0.725, 2.175), divided through by the smallest to reach the ratio 2 : 1 : 3, gave the empirical formula K_2CO_3 , computed the formula mass as $2(39) + 12 + 3(16) = 138$, found the multiplier $138 \div 138 = 1$, and concluded the molecular formula is K_2CO_3 . Every step present.

(c) Balanced equation with state symbols 0 / 1 — she wrote:



The balancing is perfect, including the leading 1. But the question asked her to "balance the following equation **and add the correct state symbols to every substance**", and there are no state symbols anywhere on the line. The mark scheme requires both, so the mark is lost. This is the sharpest single lesson on the paper: she knew the content (**she answered MCQ 19 on the meaning of (aq) correctly**) and lost the mark purely by not doing the second half of the instruction. It is the same thing that cost her a mark on Mathematics exam-15, where her algebra was right but she did not name the properties the question asked for.

Q.3(ii) Mole calculation for magnesium nitrate 5 / 5

(a) $24 + 2(14 + 48) = 24 + 124 = \mathbf{148 \text{ g/mol}}$. Correct. 1

(b) $n = 22.2 \div 148 = \mathbf{0.15 \text{ mol}}$, with the formula stated first. Correct. 2

(c) $0.15 \times 6.022 \times 10^{23} = 0.9033 \times 10^{23} = \mathbf{9.033 \times 10^{22}}$ formula units. Correct, and she handled the awkward conversion from 0.9033×10^{23} into proper standard form herself. 1

(d) **+2**, justified as:

"Mg is a group 2 metal, so it normally forms a +2 by losing it's 2 outer electrons."

Correct rule, correct value. 1

What to revise, in priority order

- **Chapter 5, chemical bonding.** This is the whole story of the paper: all four Ch5 MCQs went wrong while every other chapter held up. Specifically: a **metallic** bond is between a sea of delocalized electrons and positive metal cations (the "oppositely charged ions" wording belongs to the **ionic** bond); metals are **electropositive** because they **lose** electrons; ionic compounds have **higher** melting points than covalent ones; and atoms bond to reach the nearest **noble gas** configuration.
- **Read the second half of the instruction.** Q.3(i)(c) asked for balancing *and* state symbols. She did the balancing perfectly and stopped. Underlining the verbs in a question before answering would have earned this mark, and one on the last Maths paper too.
- **Formulae from ions.** Ca^{2+} with NO_3^- gives $\text{Ca}(\text{NO}_3)_2$. The subscript goes outside the bracket on the polyatomic ion, never on the metal.
- **Two smaller ones.** A larger electron shell means higher energy and greater distance from the nucleus, and activation energy is the minimum energy for a successful collision.

Spelling to tidy (no marks lost)

"attraction" for attraction, "acheive" for achieve, "stall" for stable, "achal" for actual. None of these cost anything here, but they are worth correcting before a board paper.

Marked 21 August 2026 from chemistry/exams/exam-19-ch1-9/attempts/2026-08-21/scans.pdf (7 pages). All 24 MCQs and every sub-part of Q.2 and Q.3 were present and attempted; no page or question was missing. Deductions logged: 7 MCQ marks + 1 mark at Q.3(i)(c) = 8, and $43 - 8 = 35$, which reconciles with the headline score.